

Swapnith Varma

Portfolio Website

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EDUCATION

- **B.Tech, Electronics and Communication Engineering**, JNTUH — CGPA: 9.05/10 2016 – 2020

SUMMARY

Software Developer with 5 years of experience building and optimizing enterprise CAD-PLM integration systems using C++, C#, and SQL. Strong CAD domain knowledge with hands-on experience in Creo Toolkit APIs and developing shared Toolkit frameworks used across multiple CAD addins. Proficient in object-oriented design and SOLID principles, with a track record of delivering significant performance improvements (70–80%). Experienced in Agile/Scrum workflows and actively upskilling in LLM internals and transformer architecture, leveraging AI tools like Amazon Q daily to accelerate and optimize development.

SKILLS

- **Programming Languages:** C++, C#, SQL, Baan 4GL, JavaScript, Python
- **Frameworks/Systems:** .NET, Infor LN, React, Node.js, REST APIs
- **Tools & IDEs:** Git, SVN, JIRA, Amazon Q, Docker, Eclipse, Visual Studio
- **Learning/Exposure:** LLMs, NLP Preprocessing, Attention Mechanisms
- **Cloud Platforms:** Infor LN Cloud (Enterprise SaaS – CAD–PLM integration)

EXPERIENCE

• Infor

Software Engineer (Jun 2024 – Present), Associate Software Engineer (Apr 2021 – May 2024)

Hyderabad, India

Enhancements in Office Integration to PLM Cloud

C#

- Built end-to-end integration between Infor PLM Cloud and Microsoft 365 using Microsoft APIs, enabling seamless saving of Word, Excel, and PowerPoint files directly from OneDrive to PLM Cloud.
- Developed a batch registration pipeline to bulk upload and manage large volumes of Office files via Microsoft APIs, significantly reducing manual effort for enterprise customers.
- Enabled multiple customers to centralize and manage their Office files in PLM Cloud, helping teams manage documents more efficiently across their organization.

Custom Defined Fields From CAD Integration

C++, C#, SQL, 4GL

- Designed a logic-driven solution to import ERP-defined custom fields into CAD Integration, enabling design-specific customizations and seamless ERP–CAD synchronization, widely used by all customers.

Performance Optimization in CAD-PLM Integration

C++, C#, SQL, 4GL

- Optimized CAD–PLM integration save operations by enhancing Toolkit and server-side logic, achieving 70–80% faster saves while ensuring reliable and stable data processing at scale in PLM Cloud.
- Designed and implemented JSON-based request handling to manage multiple interlinked database save operations and applied concurrency concepts such as multi-threading, synchronization, and request orchestration.

Creo Integration

C++

- Migrated Creo integration from v8 to v9+ by adapting to new API structures, rewriting functionalities, and validating across CAD operations, ensuring continued compatibility for enterprise customers.
- Developed top-level download functionality that replicates the full item hierarchy defined in PLM into Creo as a structured assembly, triggered by a single item ID.
- Implemented insert-replace functionality enabling users to swap an existing component in a Creo assembly with a different one fetched from PLM, maintaining assembly structure and relationships.

CAD Design Variant Handling in PLM Cloud

C++, C#, SQL, 4GL

- Enhanced Creo–PLM integration by implementing variant-aware saving functionality using STL that creates separate documents for each design variant with linked neutral files, while maintaining synchronized relationships with the generic document.
- This was a customer-critical requirement for go-live ensuring on-time implementation by overcoming major technical challenges.

Integration of Vertex, Oracle AutoVue Viewer with Infor PLM Cloud *C#, SQL, 4GL, Javascript, Java*

- Integrated Vertex CAD Viewer with PLM Cloud using REST APIs, designing a C# proof-of-concept for file uploads and retrievals, then implementing production integration in Infor LN for 2D/3D previews, markups, and seamless synchronization.
- Integrated Oracle AutoVue Desktop Viewer with PLM Cloud for file download, markup creation, and saving back via a custom button-based interface.
- Delivered seamless CAD visualization and review for on-premise and cloud customers, now actively used in production environments.

Multi Language Mapping in CAD-PLM Cloud Integration *C++, C#, SQL, 4GL*

- Implemented multi-language mapping where users could define Mapping rules in multiple languages which will save data across multiple languages in a single click.
- Actively used by customers, saving time by eliminating the need to switch languages, thereby improving efficiency.

CAD Utility from Creo *C++, C#, SQL, 4GL*

- Extended the CAD Utility to support Creo integration, enabling it to automatically retrieve records, open CAD files, generate neutral files, perform save operations, and close without any user intervention — fully automating background workflows end to end.

Export Import of Mapping Templates across Different Tenants *Infor SQL, 4GL*

- Developed a feature to export CAD Mapping Templates with all associated rules and linked tables, enabling seamless one-click import into other tenants. Eliminated the need for lengthy data migration processes, significantly reducing setup time and effort for customers across multiple environments.
- Shipped major Creo Integration features spanning CAD operations and PLM synchronization; enhanced shared Toolkit frameworks leveraged across Inventor, SolidWorks, and AutoCAD addins. Designed reusable **WPF** dialogs in C# for data display and editing, wiring user interactions to validated backend saves. Debugged and resolved complex defects critical to customer go-live stability.
- Improved robustness of shared CAD-PLM integration components by developing **MSTest unit tests** covering valid scenarios, null handling, malformed inputs, and edge cases.
- Worked in **Agile** methodology with monthly sprints and bi-annual releases, collaborating with cross-functional teams to gather requirements, track progress via Jira, and deliver customer-focused features within tight timelines.
- Assisted newly joined and existing team members in resolving issues and clarifying requirements, promoting smoother collaboration and knowledge sharing.
- Actively integrating AI tools like **Amazon Q** into day-to-day development workflows, while building foundational knowledge in **LLM** architecture — having completed data preprocessing pipelines and currently working through attention mechanisms and transformer internals

PROJECTS

Lightweight Software Updater & Release Manager *C#, GIT*

- Built a C#-based light weight auto-update system using GitHub Releases and metadata-driven version tracking, enabling fully automated patch delivery, secure ZIP extraction, and integration DLL deployment for versioned software updates.

Portfolio Website [Live Site](#) *JavaScript, React, CSS, Vite*

- Built a responsive portfolio website using React, JavaScript, and CSS. Implemented dynamic project sections with modal interactions and smooth animations, following a reusable component-based architecture with efficient state management. Optimized performance using Vite and ensured cross-device compatibility.